Skills Summary

Analyzing Transformations with Multiple Parameters

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the four parameters off a rule

Master form: $g(x) = a\,f\big(b(x-h)\big) + k$

- a: vertical stretch (reflect in the x-axis if $a < 0$) k: shift up
- b: horizontal stretch by $1/b$ (reflect in the y-axis if $b < 0$) h: shift right
- Inside changes (b, h) act *opposite* to how they look; outside changes (a, k) act as they look.

Example: $f(x) = x^2$ and $g(x) = f(x-5) + 2$. Find $g(7)$.
Step 1. Only h and k are present, so the graph shifts right 5 and up 2 — feed the shifted input to f first.
Step 2. $g(7) = f(7-5) + 2 = f(2) + 2 = 4 + 2$.
$\Rightarrow$ $\boldsymbol{g(7) = 6}$

Try it: $f(x) = x^2$ and $g(x) = f(x+3) - 1$. Find $g(1)$.

check: $\boldsymbol{g(1) = 15}$

Skill 2 — Evaluating a transformed function

Order of operations for $g(x) = a\,f(bx+c) + k$:
transform the input, apply f, then transform the output. Nothing outside f can ever change which value of f you look up.

Example: $f(x) = x^2 + 1$ and $g(x) = f(3x) - 4$. Find $g(2)$.
Step 1. The $3x$ is inside, so it changes the input before f acts; the -4 is outside and only moves the answer afterwards.
Step 2. $g(2) = f(6) - 4 = (36 + 1) - 4$.
$\Rightarrow$ $\boldsymbol{g(2) = 33}$

Try it: $f(x) = x^2 + 1$ and $h(x) = f(2x) + 5$. Find $h(3)$.

check: $\boldsymbol{h(3) = 42}$

(!) Watch out: $f(2x)$ is *not* $2f(x)$. One doubles the input, the other doubles the output — on a parabola they give different numbers at every point except the vertex.

Skill 3 — Expanding the transformed rule

To expand $g(x) = a\,f(x-h) + k$: substitute $x - h$ for *every* x in the rule for f, expand, then multiply by a and add k.

This is how you check a transformation answer: the expanded form must agree with the original at every test value.

Example: $f(x) = x^2$ and $g(x) = 3f(x-2) - 4$. Expand.
Step 1. Substitution comes first — expanding before substituting is where the middle term goes missing.
Step 2. $3(x-2)^2 - 4 = 3\left(x^2 - 4x + 4\right) - 4 = 3x^2 - 12x + 12 - 4$.
$\Rightarrow \quad \boldsymbol{g(x) = 3x^2 - 12x + 8}$

Try it: $f(x) = x^2$ and $g(x) = 2f(x+1) + 5$. Expand.

check: $\boldsymbol{g(x) = 2x^2 + 4x + 7}$